

Analysis of Psychological Stress Responses to Physiological Data from Wearable Devices: A Study of College Student Cohorts

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September 19, 2026

Abstract

Stress is an important factor in student well-being, yet it can be difficult to measure objectively using self-reported information alone. This study investigates whether physiological and behavioral data collected from Fitbit wearable devices can be used to analyze and predict the levels of mental stress and anxiety for college students. Physiological data elements include activity level, Root Mean Square of Successive Differences (RMSSD) of heart rate variability (HRV), heart rate-to-heart rate variability ratio (lf/hf), deep sleep duration, sleep quality score, step count, and oxygen saturation. Additionally, the participants provided daily ratings of stress and anxiety using a mobile application. This publicly available data was collected throughout the duration of a semester (February to July 2026) from 35 undergraduate volunteers at two universities in Mexico (<https://doi.org/10.5281/zenodo.18706837>). We examined the correlations between self-reported stress/anxiety and HRV for each individual. Correlation strength and direction varied substantially across participants. The group-level mean correlation was effectively close to zero based on this dataset. These findings indicate that the psychological stress to physiological relationship is highly idiosyncratic rather than uniform, reflecting individual differences in autonomic reactivity, baseline HRV, coping style, and behavioral expression of stress.

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1 Introduction

Welcome to your LaTeX document! This template is pre-configured with everything you need. You can write your body paragraphs smoothly. To cite a source inline, you can use the cite command like this [1].

2 Formatting Examples

Here are a few essential structural blocks to jumpstart your writing.

2.1 Mathematics and Equations

You can write inline math like this: $E = mc^2$. For standalone equations that require number labels, use the `equation` block:

$$f(x) = \int_a^b \frac{1}{x} dx \quad (1)$$

2.2 Adding Tables

In this table below, the 95% CI for Stress is between stress and average RMSSD. All ρ values (Spearman's rank correlation coefficient) are between the respective coefficient and average RMSSD. Note that some participants (out of 35) have been removed due to lack of data.

Table 1: A table of correlations across the participants.

Participant	Number	RMSSD	95% CI for Stress	ρ stress	ρ anxiety
1	27	69.260	[-0.2, 0.53]	0.148	0.191
2	06	29.346	[-0.14, 0.97]	0.519	0.757
4	06	39.121	[-0.84, 0.77]	0.007	-0.105
5	85	27.947	[0.02, 0.43]	0.200	0.234
6	32	103.713	[-0.26, 0.44]	0.080	0.103
7	83	126.437	[-0.1, 0.33]	0.181	0.120
8	89	54.964	[0.04, 0.43]	0.242	0.244
10	34	95.730	[-0.2, 0.46]	0.064	0.148
17	05	43.132	[0.62, 1.0]	0.996	0.971
19	41	105.997	[-0.52, 0.06]	0.064	-0.255
20	38	60.337	[0.08, 0.63]	0.044	0.390
22	19	111.330	[-0.19, 0.66]	0.046	0.288
24	09	107.419	[-0.31, 0.06]	-0.294	-0.130
26	68	67.118	[-0.04, 0.42]	0.086	0.204
27	84	69.800	[-0.4, 0.01]	-0.144	-0.204
28	35	26.514	[-0.52, 0.12]	-0.382	-0.225
29	28	62.548	[-0.25, 0.48]	0.160	0.134
30	75	40.945	[-0.33, 0.12]	0.155	-0.106
32	71	71.783	[-0.53, -0.12]	-0.249	-0.340
33	49	64.126	[-0.15, 0.4]	0.152	0.139
34	51	63.114	[-0.28, 0.27]	-0.134	-0.010

2.3 Inserting Figures

To insert an image file, ensure the file (e.g., `chart.png`) is uploaded to your working directory and use the code below:

Figure 1: Placeholder caption for your visual data.

3 Conclusion

This concludes the default structural blueprint of the document.

References

- [1] Author, A. (2026). *The Title of the Authored Work*. Journal of Examples, 12(3), 45-67.